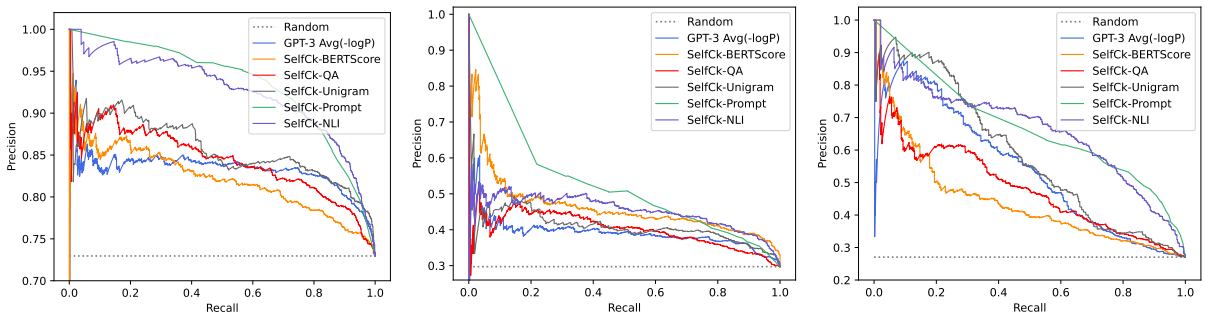


(a) Non-Factual Sentences **(b)** Non-Factual* Sentences **(c)** Factual Sentences

Figure 5: PR-Curve of detecting non-factual and factual *sentences* in the GPT-3 generated WikiBio passages.

Method	Sentence-level (AUC-PR)			Passage-level (Corr.)	
	NonFact	NonFact*	Factual	Pearson	Spearman
Random	72.96	29.72	27.04	-	-
`GPT-3 (text-davinci-003)`'s probabilities (*LLM, grey-box*)					
Avg($-\log p$)	83.21	38.89	53.97	57.04	53.93
Avg($\mathcal{H}$)†	80.73	37.09	52.07	55.52	50.87
Max($-\log p$)	87.51	35.88	50.46	57.83	55.69
Max($\mathcal{H}$)†	85.75	32.43	50.27	52.48	49.55
`LLaMA-30B`'s probabilities (*Proxy LLM, black-box*)					
Avg($-\log p$)	75.43	30.32	41.29	21.72	20.20
Avg($\mathcal{H}$)	80.80	39.01	42.97	33.80	39.49
Max($-\log p$)	74.01	27.14	31.08	-22.83	-22.71
Max($\mathcal{H}$)	80.92	37.32	37.90	35.57	38.94
SelfCheckGPT (*black-box*)					
w/ BERTScore	81.96	45.96	44.23	58.18	55.90
w/ QA	84.26	40.06	48.14	61.07	59.29
w/ Unigram (max)	85.63	41.04	58.47	64.71	64.91
w/ NLI	92.50	45.17	66.08	74.14	73.78
w/ Prompt	**93.42**	**53.19**	**67.09**	**78.32**	**78.30**

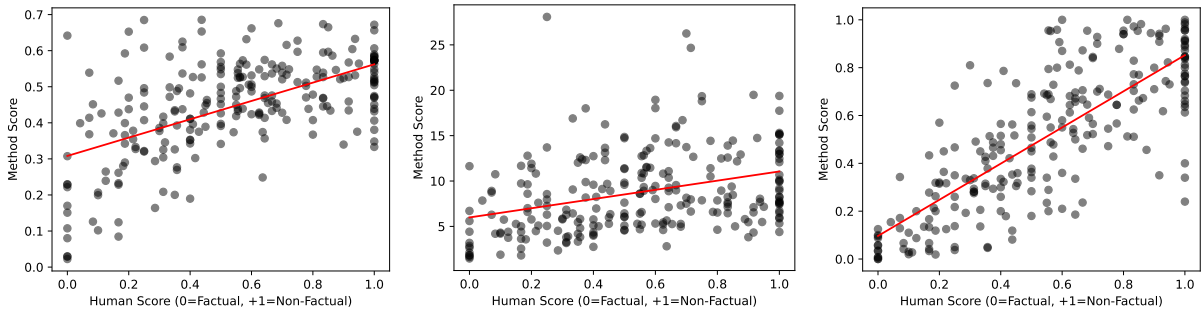
Table 2: AUC-PR for sentence-level detection tasks. Passage-level ranking performances are measured by Pearson correlation coefficient and Spearman's rank correlation coefficient w.r.t. human judgements. The results of other proxy LLMs, in addition to LLaMA, can be found in the appendix. †GPT-3 API returns the top-5 tokens' probabilities, which are used to compute entropy.

low probability in situations where the response is dissimilar to the generation style of the proxy LLM. We note that a weighted conditional LM score such as BARTScore (Yuan et al., 2021) could be incorporated in future investigations.

3) SelfCheckGPT outperforms grey-box approaches. It can be seen that SelfCheckGPT-Prompt *considerably* outperforms the grey-box approaches (including GPT-3's output probabilities) as well as other black-box approaches. Even other variants of SelfCheckGPT, including BERTScore, QA, and n-gram, outperform the grey-box approaches in most setups. Interestingly, despite being the least computationally expensive method, SelfCheckGPT with unigram (max) works well across different setups. Essentially, when assessing a sentence, this method picks up the token with the *lowest* occurrence given all the samples. This suggests that if a token only appears a few times (or once) within the generated samples (N=20), it is likely non-factual.

4) SelfCheckGPT with n-gram. When investigating the n-gram performance from 1-gram to 5-gram, the results show that simply finding the least likely token/n-gram is more effective than computing the average n-gram score of the sentence, details in appendix Table 7. Additionally, as n increases, the performance of SelfCheckGPT with n-gram (max) drops.

5) SelfCheckGPT with NLI. The NLI-based



(a) GPT-3 Avg($-\log p$) **(b)** LLaMA-30B Avg($\mathcal{H}$) **(c)** SelfCheckGPT-Prompt

Figure 6: Scatter plot of passage-level scores where Y-axis = Method scores, X-axis = Human scores. Correlations are reported in Table 2. The scatter plots of other SelfCheckGPT variants are provided in Figure 10 in the appendix.

method outperforms all black-box and grey-box baselines, and its performance is close to the performance of the Prompt method. As SelfCheckGPT with Prompt can be computationally heavy, SelfCheckGPT with NLI could be the most practical method as it provides a good trade-off between performance and computation.

7.2 Passage-level Factuality Ranking

Previous results demonstrate that SelfCheckGPT is an effective approach for predicting sentence-level factuality. An additional consideration is whether SelfCheckGPT can also be used to determine the overall factuality of passages. Passage-level factuality scores are calculated by averaging the sentence-level scores over all sentences.

$$\mathcal{S}_{\text{passage}} = \frac{1}{|R|} \sum_{i} \mathcal{S}(i) \tag{12}$$

where $\mathcal{S}(i)$ is the sentence-level score, and $|R|$ is the number of sentences in the passage. Since human judgement is somewhat subjective, averaging the sentence-level labels would lead to ground truths with less noise. Note that for Avg($-\log p$) and Avg($\mathcal{H}$), we compute the average over all tokens in a passage. Whereas for Max($-\log p$) and Max($\mathcal{H}$), we first take the maximum operation over tokens at the sentence level, and we then average over all sentences following Equation 12.

Our results in Table 2 and Figure 6 show that all SelfCheckGPT methods correlate far better with human judgements than the other baselines, including the grey-box probability and entropy methods. SelfCheckGPT-Prompt is the best-performing method, achieving the highest Pearson correlation of 78.32. Unsurprisingly, the proxy LLM approach again achieves considerably lower correlations.

7.3 Ablation Studies

External Knowledge (instead of SelfCheck)

If external knowledge is available, one can measure the informational consistency between the LLM response and the information source. In this experiment, we use the first paragraph of each concept that is available in WikiBio.[6]

Method	Sent-lvl AUC-PR			Passage-lvl	
	NoFac	NoFac*	Fact	Pear.	Spear.
SelfCk-BERT	81.96	45.96	44.23	58.18	55.90
WikiBio+BERT	81.32	40.62	49.15	58.71	55.80
SelfCk-QA	84.26	40.06	48.14	61.07	59.29
WikiBio+QA	84.18	45.40	52.03	57.26	53.62
SelfCk-1gm	85.63	41.04	58.47	64.71	64.91
WikiBio+1gm	80.43	31.47	40.53	28.67	26.70
SelfCk-NLI	92.50	45.17	66.08	74.14	73.78
WikiBio+NLI	91.18	48.14	71.61	78.84	80.00
SelfCk-Prompt	93.42	53.19	67.09	78.32	78.30
WikiBio+Prompt	93.59	65.26	73.11	85.90	86.11

Table 3: The performance when using SelfCheckGPT samples versus external stored knowledge.

Our findings in Table 3 show the following. First, SelfCheckGPT with BERTScore/QA, using self-samples, can yield comparable or even better performance than when using the reference passage. Second, SelfCheckGPT with n-gram shows a large performance drop when using the WikiBio passages instead of self-samples. This failure is attributed to the fact that the WikiBio reference text alone is not sufficient to train an n-gram model. Third, in contrast, SelfCheckGPT with NLI/Prompt can benefit considerably when access to retrieved information is available. Nevertheless, in practice,

[6]This method is no longer zero-resource as it requires retrieving relevant knowledge from external data.

it is infeasible to have an external database for every possible use case of LLM generation.

The Impact of the Number of Samples

Although sample-based methods are expected to perform better when more samples are drawn, this has higher computational costs. Thus, we investigate performance as the number of samples is varied. Our results in Figure 7 show that the performance of SelfCheckGPT increases smoothly as more samples are used, with diminishing gains as more samples are generated. SelfCheckGPT with n-gram requires the highest number of samples before its performance reaches a plateau.

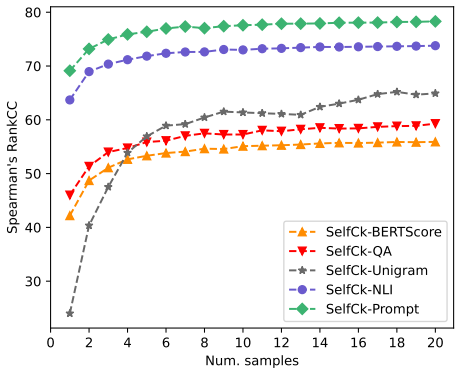


Figure 7: The performance of SelfCheckGPT methods on ranking passages (Spearman's) versus the number of samples.

The Choice of LLM for SelfCheckGPT-Prompt

We investigate whether the LLM generating the text can self-check its own text. We conduct this ablation using a reduced set of the samples (N=4).

Text-Gen	SelfCk-Prompt	N	Pear.	Spear.
GPT-3	ChatGPT	20	78.32	78.30
GPT-3	ChatGPT	4	76.47	76.41
GPT-3	GPT-3	4	73.11	74.69
†SelfCheck w/ unigram (max)		20	64.71	64.91
†SelfCheck w/ NLI		20	74.14	73.78

Table 4: Comparison of GPT-3 (text-davinci-003) and ChatGPT (gpt-3.5.turbo) as the prompt-based text evaluator in SelfCheckGPT-Prompt. †Taken from Table 2 for comparison.

The results in Table 4 show that GPT-3 can self-check its own text, and is better than the unigram method even when using only 4 samples. However, ChatGPT shows a slight improvement over GPT-3 in evaluating whether the sentence is supported by the context. More details are in Appendix C.

8 Conclusions

This paper is the first work to consider the task of hallucination detection for general large language model responses. We propose SelfCheckGPT, a zero-resource approach that is applicable to any black-box LLM without the need for external resources, and demonstrate the efficacy of our method. SelfCheckGPT outperforms a range of considered grey-box and black-box baseline detection methods at both the sentence and passage levels, and we further release an annotated dataset for GPT-3 hallucination detection with sentence-level factuality labels.

Limitations

In this study, the 238 GPT-3 generated texts were predominantly passages about individuals in the WikiBio dataset. To further investigate the nature of LLM's hallucination, this study could be extended to a wider range of concepts, e.g., to also consider generated texts about locations and objects. Further, this work considers factuality at the sentence level, but we note that a single sentence may consist of both factual and non-factual information. For example, the following work by Min et al. (2023) considers a fine-grained factuality evaluation by decomposing sentences into atomic facts. Finally, SelfCheckGPT with Prompt, which was convincingly the best selfcheck method, is quite computationally heavy. This might lead to impractical computational costs, which could be addressed in future work to be made more efficient.

Ethics Statement

As this work addresses the issue of LLM's hallucination, we note that if hallucinated contents are not detected, they could lead to misinformation.

Acknowledgments

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